

Rebuttal to Reviewer hyVA

We sincerely thank the reviewer for the thorough evaluation of our manuscript and for the constructive and valuable feedback. We greatly appreciate the reviewer's comments and suggestions, which have helped us identify areas requiring clarification and improve the overall quality and presentation of our work.

We have carefully addressed each of the reviewer's comments below. We have provided detailed explanations and, where appropriate, additional experimental details and results to clarify the methodology and findings.

Q1. Video length, fps, BiLSTM sequence length, and the temporal window needed.

The videos were recorded at approximately 30 fps. As an example, the Sabai Vanakkam sequence contains 860 frames, corresponding to approximately 28.7 seconds. The videos have varying durations rather than a single fixed sequence length. The Conv1D-BiLSTM processes the frame-level feature sequences and provides temporal context in both forward and backward directions within the input sequence.

We did not perform a temporal-window ablation in the current study. Therefore, we cannot specify the minimum temporal window required for reliable Labanotation prediction. We acknowledge that systematically evaluating different temporal context lengths would be valuable for determining how much movement history is necessary for different notation attributes.

Q2. How were the 23 features determined? References? Meaningful from a dance perspective?

The 23 values are the per-frame input to the sequence model (Sec. 4.8): 8 joint angles (elbow, shoulder, hip, knee, left and right) and 15 movement descriptors. The effort, shape and space features follow Laban Movement Analysis concepts [10, 11]. The rest are standard kinematic and rhythm descriptors. The set was chosen by the authors to cover the notation columns being predicted. We ran no feature ablation and no expert review, so the features are motivated by LMA but their dance meaning has not been validated.

Q3. How is foreshortening handled without 3D?

It is not. Normalisation by torso length reduces scale and camera-distance effects but does not correct foreshortening. This is why arm direction and foot contact stay at about 53% and 47–54% (Table 7). A calibrated multi-view setup, as the reviewer suggests, would be the principled remedy.

Q4. Basis for the weights in Section 4.4.

We acknowledge that the weights used in Section 4.4 were hand-set and that no formal sensitivity analysis or optimization was performed. The four weights—velocity (0.35), acceleration (0.25), reach (0.20), and contraction (0.20)—were chosen heuristically to favour temporally stable and clearly held postures during keyframe selection. Accordingly, velocity and acceleration together receive a higher weight (0.60) than reach and contraction (0.40), based on the observation that clearer, more stable poses are preferable to frames captured during transitional or mid-swing movements.

These weights were informed by manual inspection of keyframe quality on trial videos rather than by a formal grid search or data-driven optimization. We agree that this design choice was not sufficiently explained in the manuscript. We also acknowledge that the selected weights are not necessarily optimal and may not generalize across different dance styles, performers, or tempos. A systematic sensitivity analysis or data-driven weight optimization would provide stronger validation of the posture-selection strategy and is an important direction for future work.

Q5. Is 96 videos too few for a BiLSTM?

It is small, and the 30.2% adavu result is consistent with that, though we have not tested the cause. For per-frame notation the supervision is dense, and we used dropout, label smoothing, augmentation and dancer-disjoint CV. High scores on deterministic columns do not show generalisation beyond this dataset.

We sincerely thank the reviewer for the thoughtful and constructive feedback. We hope that the responses and clarifications provided above adequately address the concerns raised.